

Pre-eclampsia, Eclampsia, & Gestational Diabetes Mellitus (GDM)

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Pre-eclampsia: Definition & Pathophysiology

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- New-onset hypertension ($\geq 140/90$) after 20 weeks + proteinuria or organ dysfunction.
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- Placental abnormal trophoblast invasion $\rightarrow$ endothelial dysfunction.
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- Increased anti-angiogenic factors (sFlt-1), decreased PlGF.
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- Systemic vasoconstriction $\rightarrow$ multi-organ involvement.

Clinical Features

- Hypertension, proteinuria

- Headache, visual disturbances

- RUQ/epigastric pain (liver involvement)

- Edema, oliguria, hyperreflexia

- Severe features: BP $\geq 160/110$, thrombocytopenia, $\uparrow$ creatinine

Pre-eclampsia: Management

- Definitive management: delivery.

- BP control: labetalol, hydralazine, nifedipine.

- Seizure prophylaxis: magnesium sulfate.

- Fetal monitoring and timing of delivery depending on severity and GA.

Eclampsia: Definition

Onset of **generalized tonic-clonic seizures** in a woman with **preeclampsia** (hypertension + proteinuria)

May occur **before, during, or after delivery**

Life-threatening maternal and fetal complication

Endothelial dysfunction → vasospasm

Hypertension → cerebral edema

Seizure activity due to cerebral ischemia and excitatory neurotransmitter imbalance

Multi-organ involvement: kidneys, liver, CNS

- **Eclampsia: General Management Principles**

Stabilize mother (airway, breathing, circulation)

Control seizures → prevent recurrence

Control blood pressure → prevent stroke or organ damage

Definitive treatment → delivery of fetus and placenta

Monitor maternal & fetal status

**Magnesium
Sulfate
(MgSO₄)**

Drug of choice
prevention
treatment
eclampsia

Mechanism: CNS
depressant → blocks
NMDA receptors, reduces
neuromuscular excitability

- **Eclampsia: First-Line
Therapy for
Seizure
Control**

Regimens:
• **Pritchard
(IM + IV)**

**Zus
(IV o**

• Eclampsia: Magnesium Sulfate Dosing

Loading dose: 4–6 g IV over 15–20 min or 10 g IM (5 g per buttock)

Maintenance dose: 1 g/hour IV or 5 g IM every 4 hours

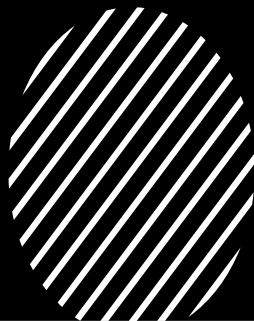
Maximum safe dose: 30–40 g/24 h

Monitoring:

- Deep tendon reflexes
- Respiratory rate >12/min
- Urine output >25–30 mL/hr



Eclampsia: Blood Pressure Management



Target: SBP <160 mmHg, DBP <110 mmHg

- **Hydralazine** (IV)
- **Labetalol** (IV)
- **Nifedipine** (oral, if stable)

First-line agents:

Avoid: ACE inhibitors, ARBs, atenolol

**Eclampsia: Other
Adjunct
Therapies**

Sedatives: rarely, only if
 MgSO_4 inadequate

Corticosteroids: for fetal
lung maturity if preterm

Diuretics: rarely indicated; may
worsen placental perfusion

Eclampsia: Monitoring

Vital signs: BP, HR, RR

Neurological status: consciousness, reflexes

Urine output: renal function

Serum magnesium levels: if prolonged therapy

Fetal monitoring: continuous if antepartum

**Eclampsia: Complications
to Watch**

Magnesium toxicity → absent reflexes,
respiratory depression, cardiac arrhythmia

Pulmonary edema

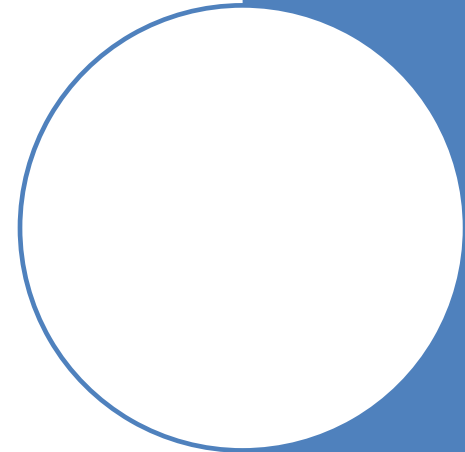
HELLP syndrome

Placental abruption

Maternal/fetal death

GDM: Definition & Pathophysiology

- Glucose intolerance first recognized during pregnancy.
- Placental hormones → insulin resistance.
- Beta-cell insufficiency → hyperglycemia.
- Risk factors: obesity, prior GDM, family history.



Screening & Diagnosis

- Screening at 24–28 weeks.

- One-step (75 g OGTT): Fasting ≥ 92 , 1h ≥ 180 , 2h ≥ 153 mg/dL $\rightarrow$ diagnosis.

- Two-step method (50 g GCT $\rightarrow$ 100 g OGTT).

GDM Management

- Lifestyle therapy: diet + moderate exercise.
- Glucose monitoring: fasting <95 mg/dL, 1-hour post-meal <140.
- Pharmacologic therapy: insulin (preferred), metformin if needed.
- Fetal surveillance: growth scans.

Maternal & Fetal Complications

- Pre-eclampsia: eclampsia, HELLP, organ failure.

- GDM: macrosomia, shoulder dystocia, neonatal hypoglycemia.

- Long-term: increased risk of type 2 diabetes.

Summary

- Pre-eclampsia = hypertensive disorder with endothelial dysfunction.

- GDM = pregnancy-induced insulin resistance and hyperglycemia.

- Early detection and appropriate management reduce complications.

Summary...

Eclampsia is a **life-threatening complication** of pregnancy

Magnesium sulfate is the **drug of choice** for seizure prevention and treatment

Blood pressure control is essential

Delivery is definitive treatment

Careful **monitoring** reduces morbidity and mortality

References

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Questions?